

CHIA-OpenROAD: Surrogate-Guided Agentic RTL-to-GDS

Barsat Khadka | University of Southern Mississippi

Problem tracks: Agent-in-the-loop RTL-to-GDS flows; compatibility for new platforms in CHIA

OVERVIEW

Agentic hardware-design systems require fast feedback during exploration and trustworthy verification before accepting a result. CHIA provides infrastructure for constructing and executing such workflows, but it does not currently include a first-class, stage-addressable integration with OpenROAD-flow-scripts (ORFS). Its existing VLSI examples primarily use Hammer and commercial synthesis tools.

I propose to add OpenROAD as a reusable execution and evaluation backend within CHIA. The resulting loop will allow an agent to inspect physical-design reports, explore bounded configuration changes, recover from failed stages, and produce a DRC/LVS-clean layout through an open-source flow. The integration will also provide a general interface for inexpensive learned surrogate models. During CTS exploration, I will use my previous work, [SwiftCTS](#), as a demonstration of this interface.

METHODOLOGY

I will implement containerized CHIA nodes for preparing an ORFS design and executing synthesis, floorplanning, placement, clock-tree synthesis, routing, and physical verification. Each node will expose its resource requirements through @ChiaFunction and return structured metrics, reports, and design checkpoints. CHIA's asynchronous execution will allow candidates to run in parallel, while caching and bypassing will allow experiments to resume from the most recent valid stage.

The workflow will separate agentic decisions from trusted evaluation. The agent will receive tools for inspecting timing, congestion, clock-tree, power, area, and violation summaries. It may select from a bounded set of legal configuration changes, launch candidate runs, compare results, and choose the next action. Execution, report parsing, constraint checking, DRC, and LVS will remain programmatic and inaccessible for modification by the agent.

During CTS exploration, the loop will invoke SwiftCTS to provide inexpensive estimates of which configurations are promising. SwiftCTS predicts clock power, clock wirelength, and timing skew using lightweight, physics-informed models. It can adapt to a new placement using one or two OpenROAD calibration runs and evaluate 100,000 CTS configurations in under ten seconds. The agent will use these predictions to select a small, diverse set of Pareto candidates for actual OpenROAD CTS and routing. OpenROAD results will remain the ground truth used to accept or reject configurations.

SwiftCTS will be integrated as a replaceable CHIA evaluator rather than being embedded directly in the agent. This interface will allow future users to insert learned models for timing, congestion, routing, or power estimation and extend the same surrogate-guided approach beyond CTS to other stages of physical design, such as floorplanning, placement, and routing, as needed.

The loop will be evaluated on a small and a medium open-source ORFS-compatible design using an open PDK. Four configurations will be compared under similar compute budgets: default ORFS, ORFS AutoTuner, the report-aware agent without surrogate screening, and the complete agent with SwiftCTS. Measurements will include tool invocations, wall-clock time, final quality of results, and DRC/LVS status.

EXPECTED RESULTS

1. Reusable, documented CHIA nodes and MCP tools for OpenROAD/ORFS.
2. An agentic RTL-to-GDS demonstration producing at least one DRC/LVS-clean layout.
3. A general interface for incorporating learned surrogate evaluators into CHIA.
4. A quantitative evaluation of the value of surrogate models within agentic physical-design loops.
5. An upstream-ready implementation and reproducible experiment configuration.

FUTURE WORK

Future work will extend the surrogate interface across additional physical-design stages and compose multiple low-cost evaluators within a single RTL-to-GDS loop. OpenROAD results collected by CHIA could also support continual calibration across new designs and PDKs. In the longer term, this could enable agents to learn when a surrogate is sufficient and when an expensive physical-design tool must be invoked.

COST ESTIMATE

Parallel CPU instances for OpenROAD experiments	\$450
Gemini model and API usage	\$180
Cloud storage for checkpoints and artifacts	\$30
Additional runs and failure contingency	\$90
Total estimated maximum cost	\$750